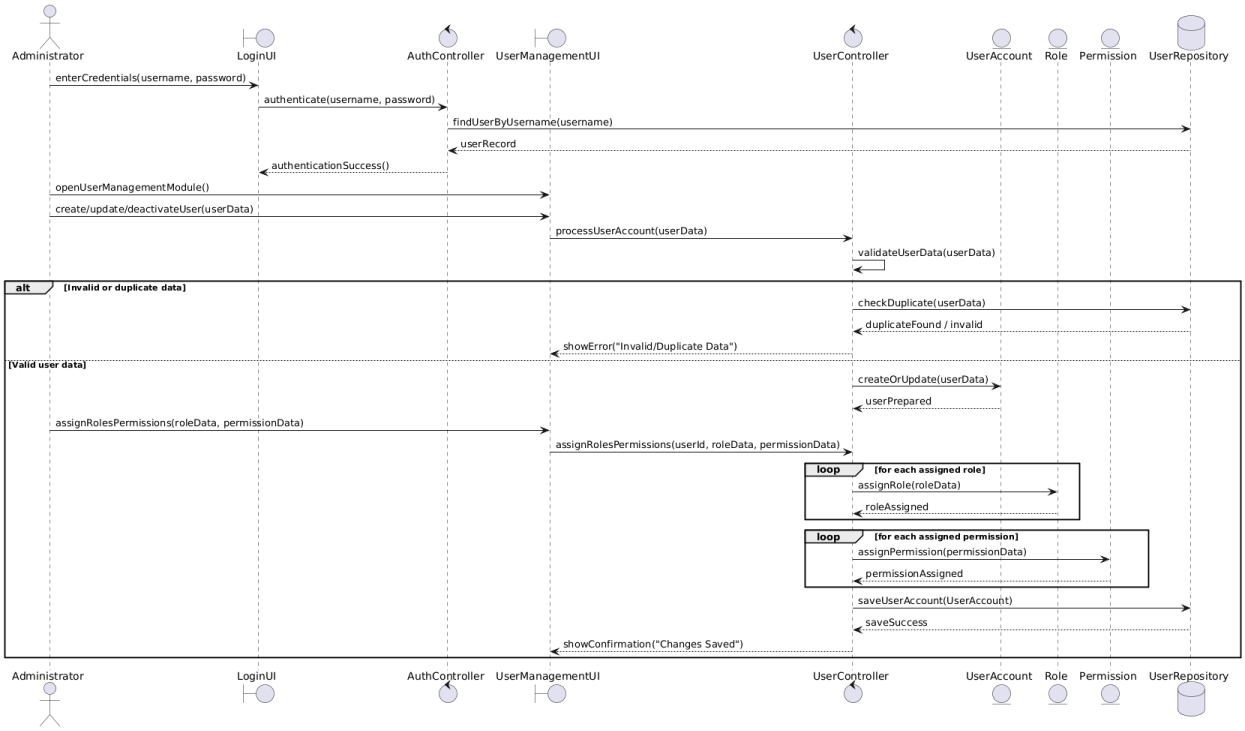
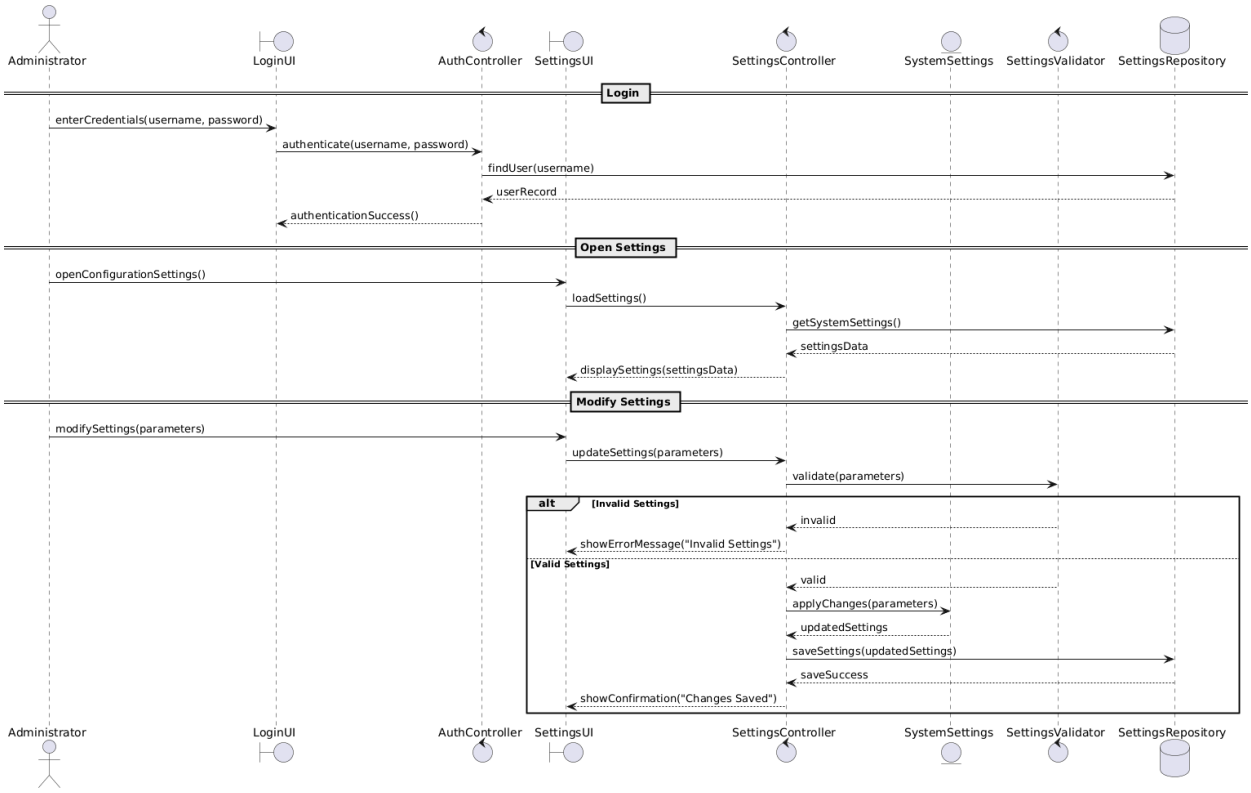
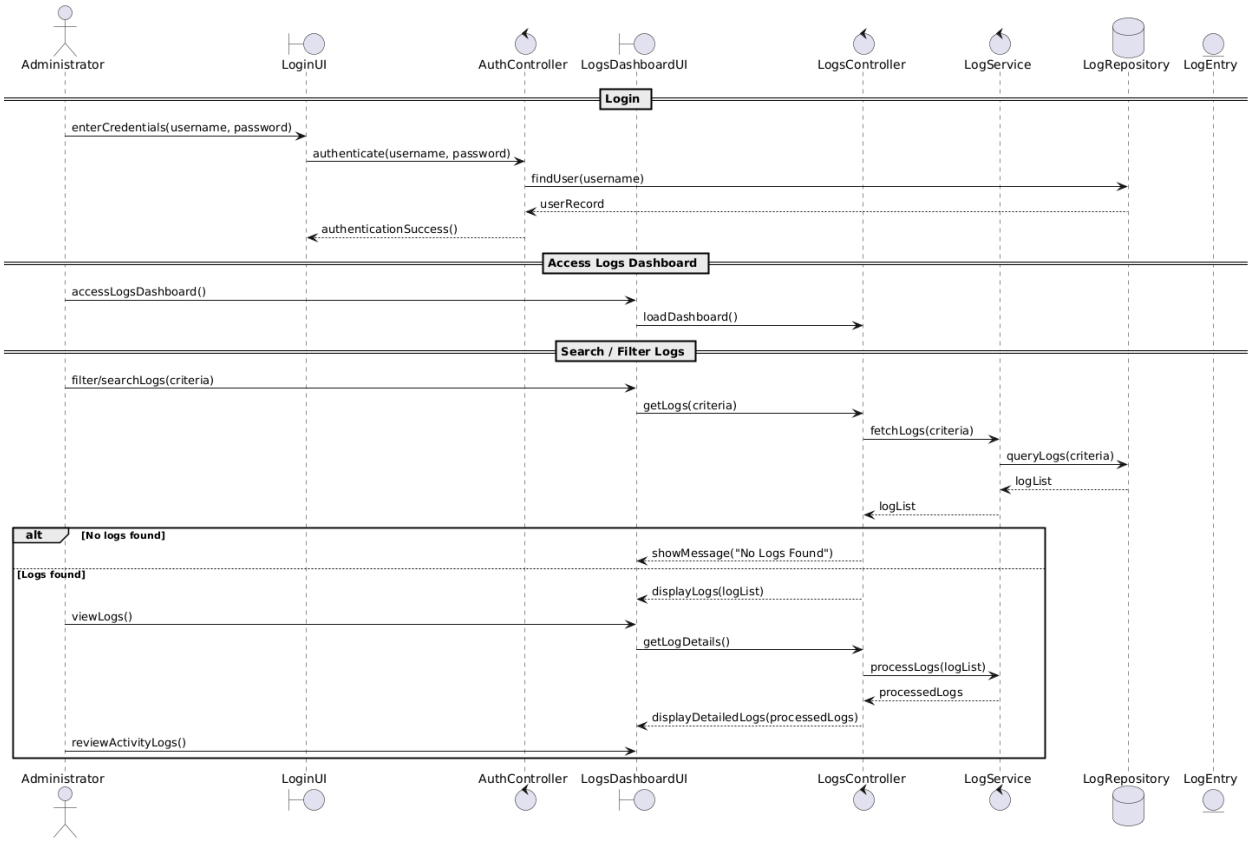
Sequence Diagram UC-01



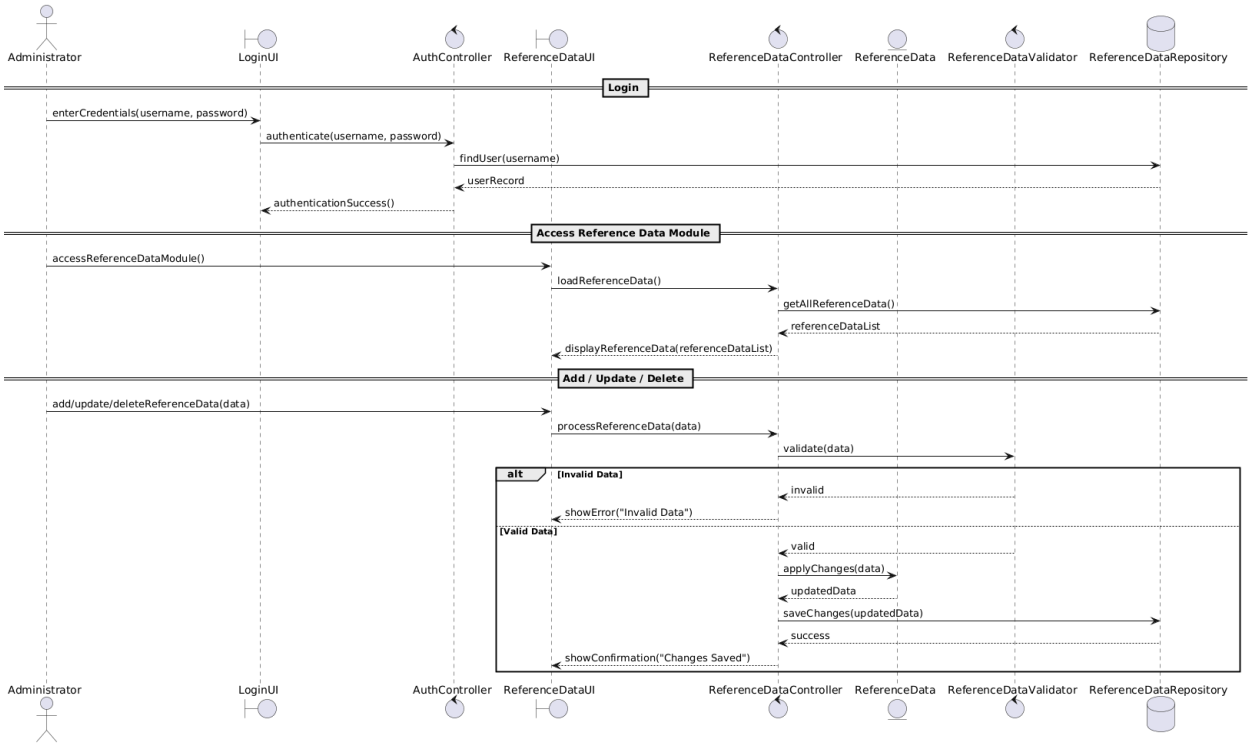
Sequence Diagram UC-02



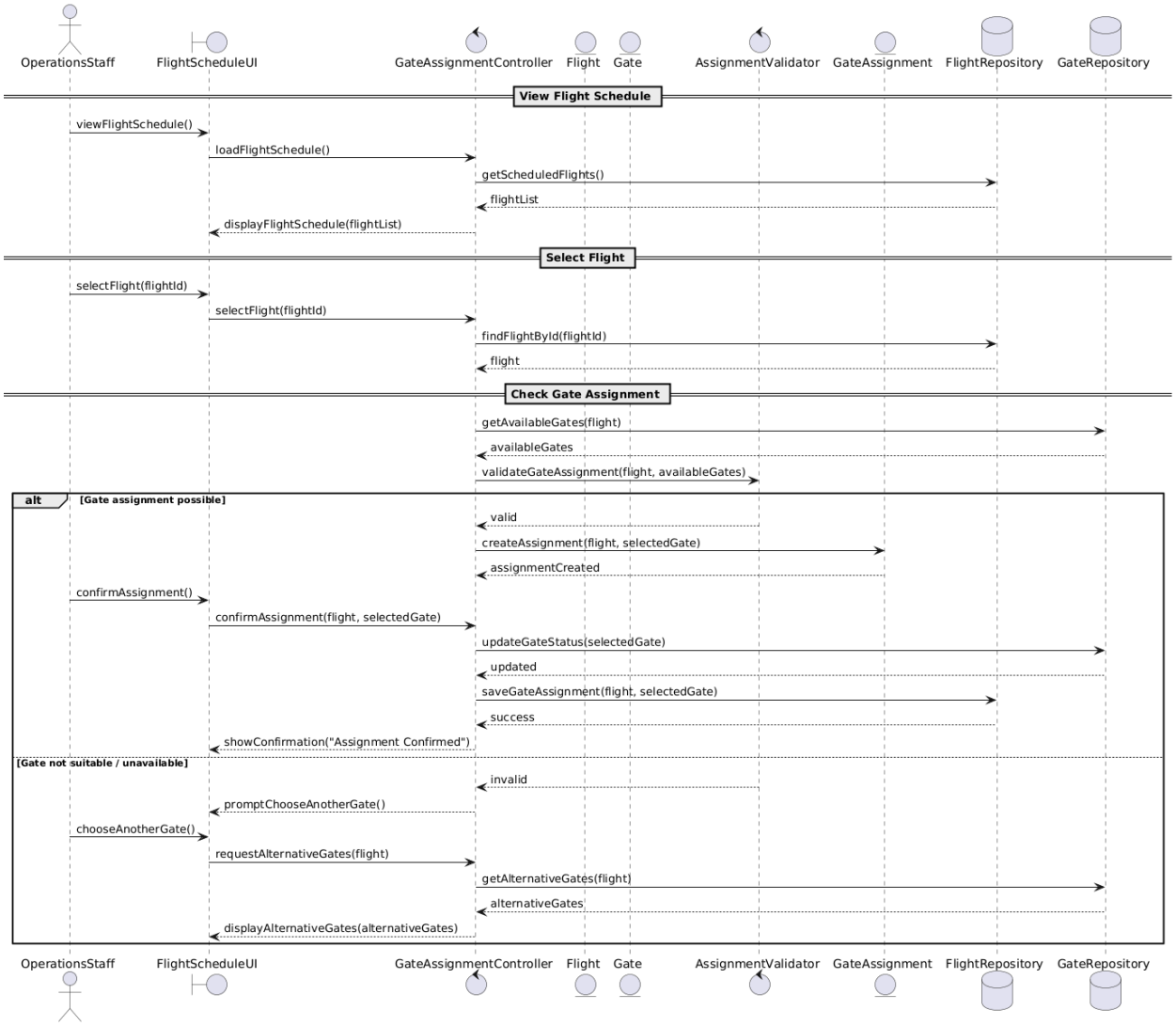
Sequence Diagram UC-03

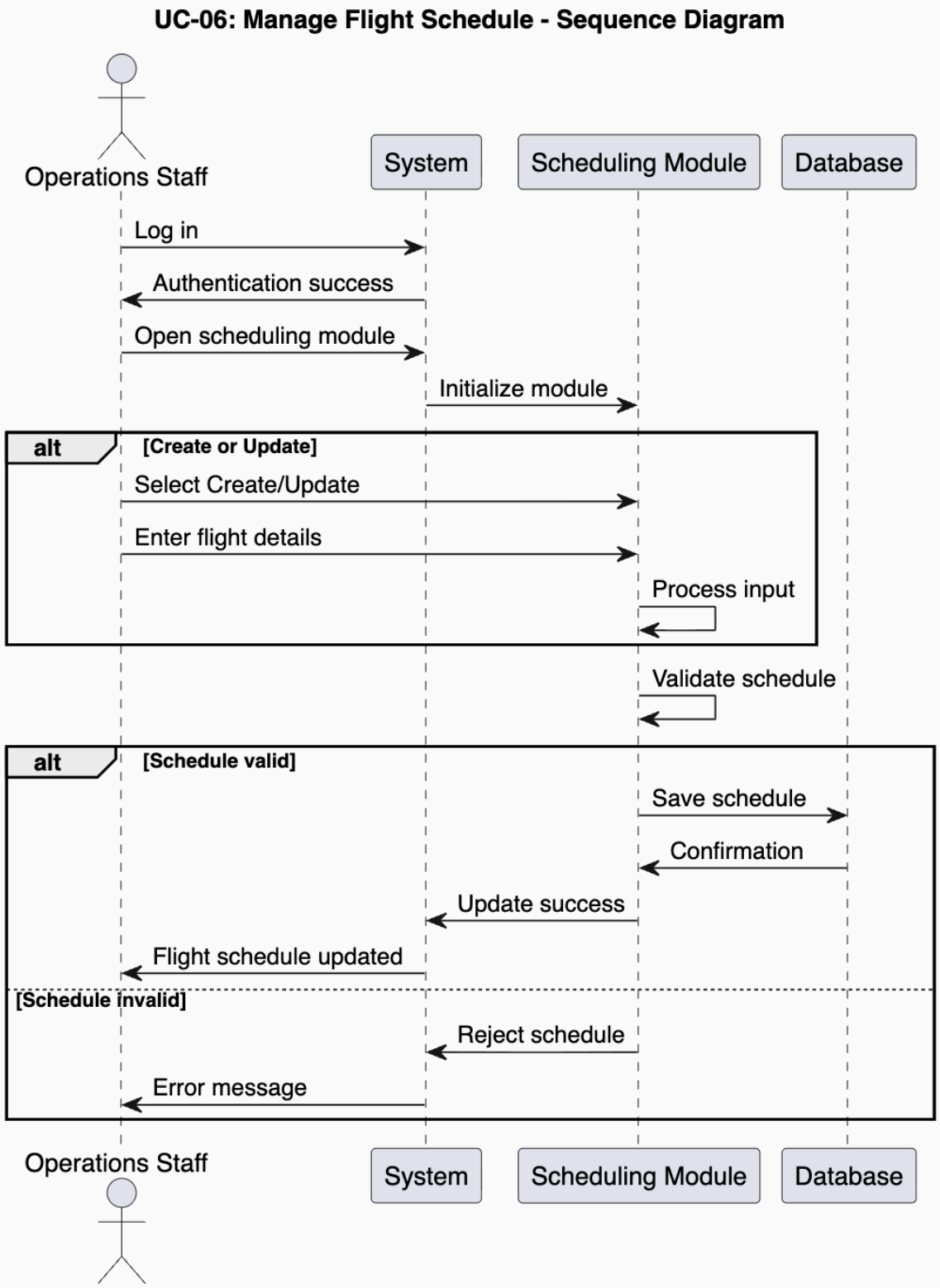


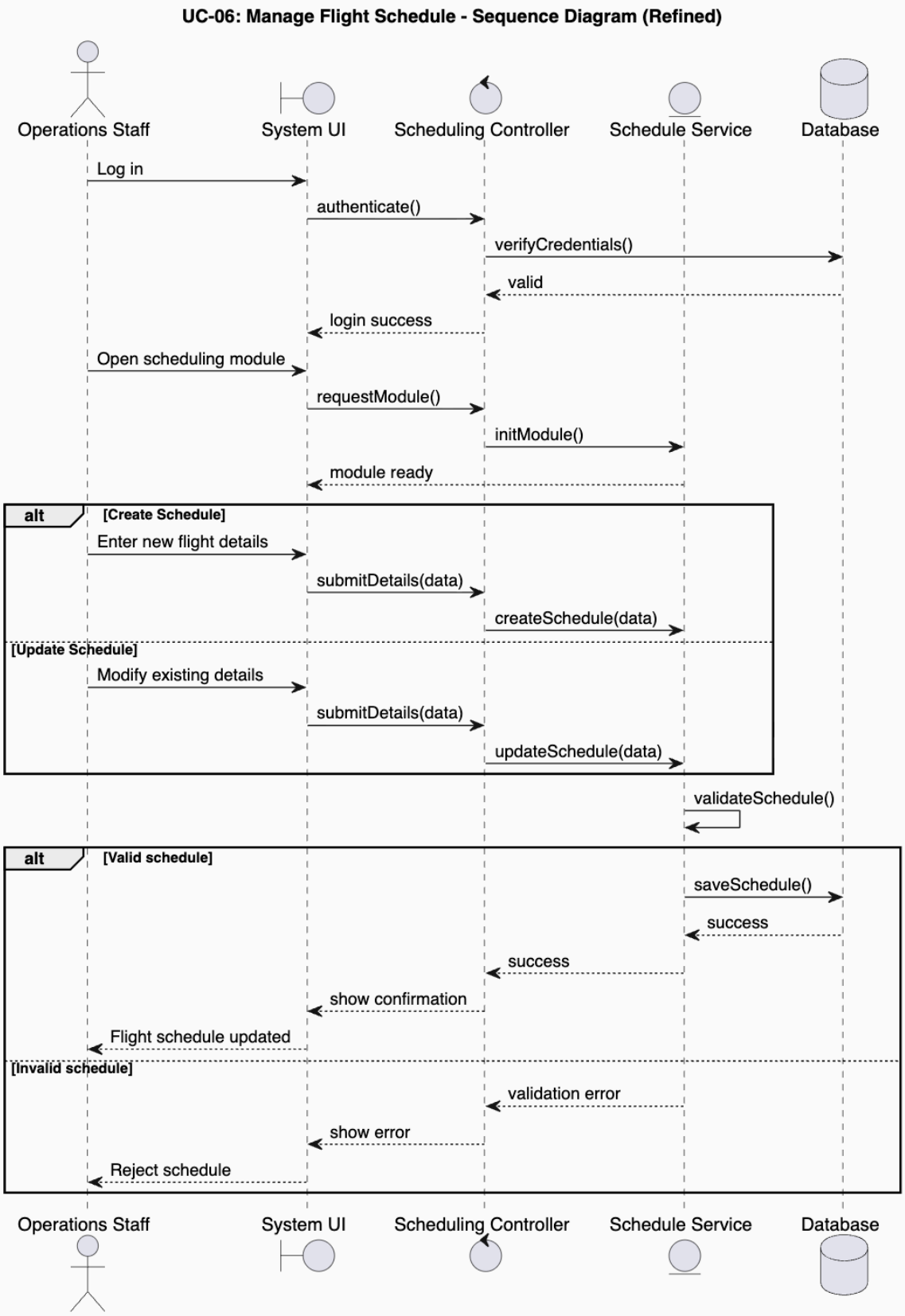
Sequence Diagram UC-04

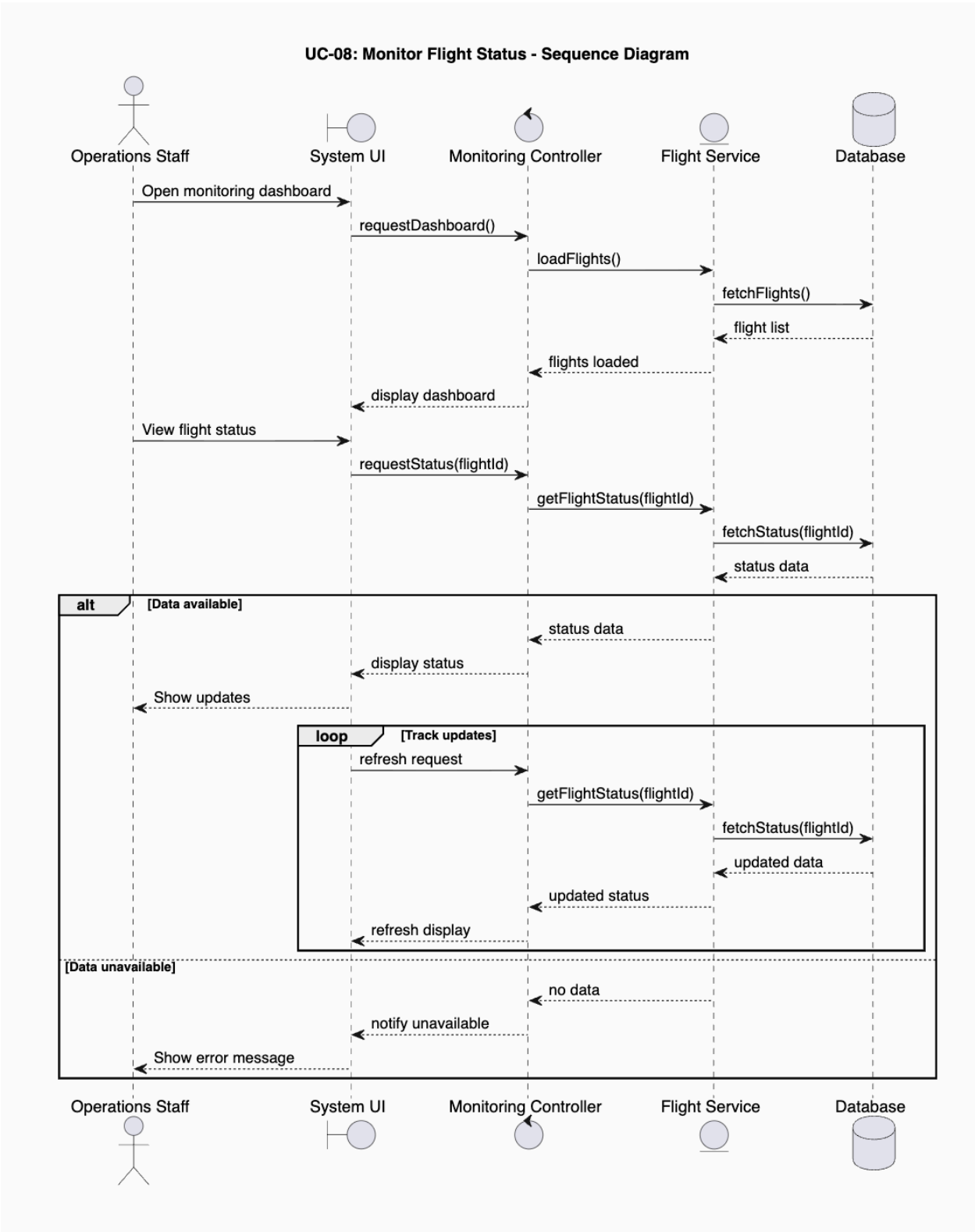


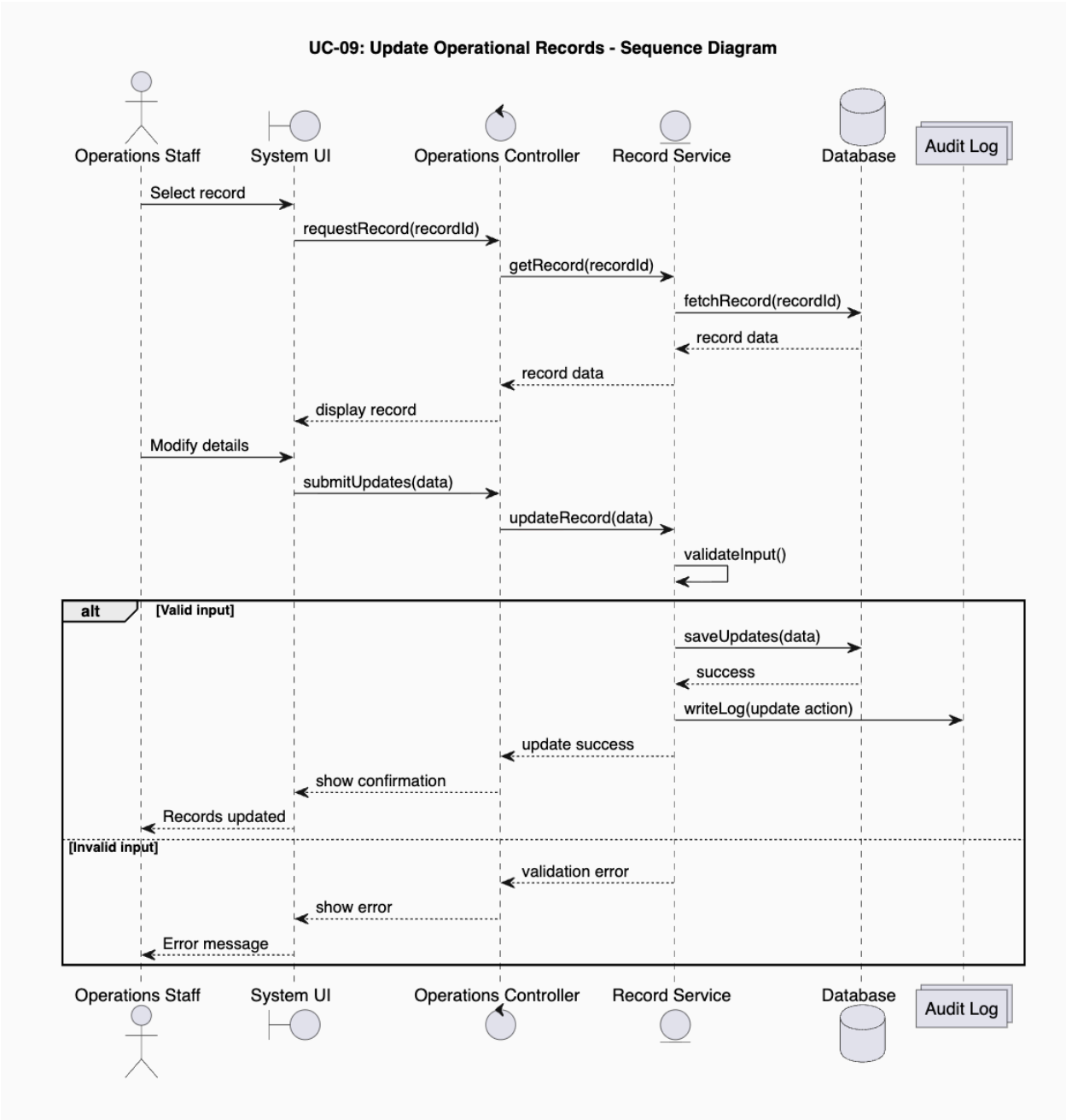
Sequence Diagram UC-05

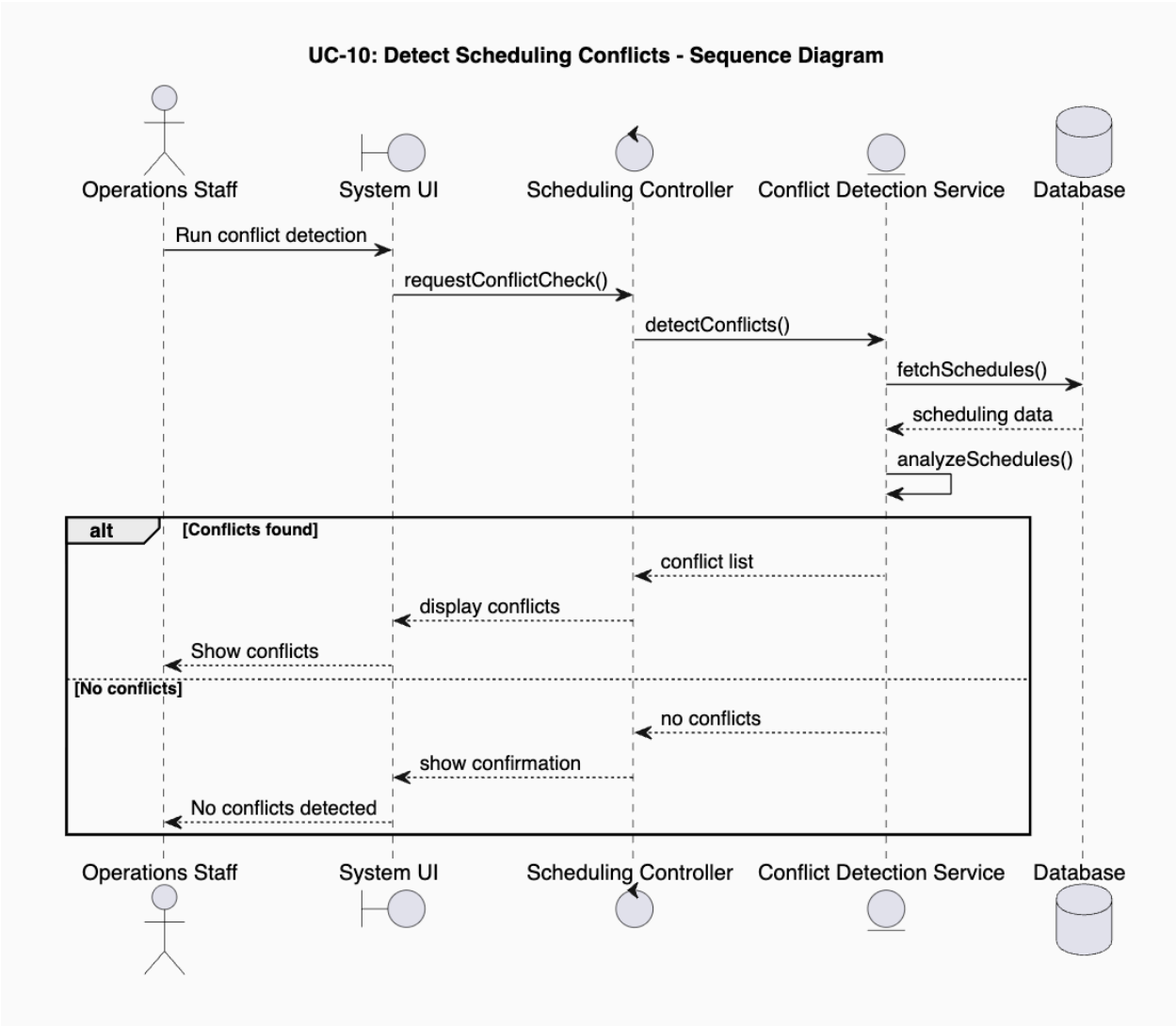



UC-06: Manage Flight Schedule - Sequence Diagram
Operations Staff
System
Scheduling Module
Database
Log in
Authentication success
Open scheduling module
Initialize module
alt
[Create or Update]
Select Create/Update
Enter flight details
Process input
Validate schedule
alt
[Schedule valid]
Save schedule
Confirmation
Update success
Flight schedule updated
[Schedule invalid]
Reject schedule
Error message
Operations Staff
System
Scheduling Module
Database

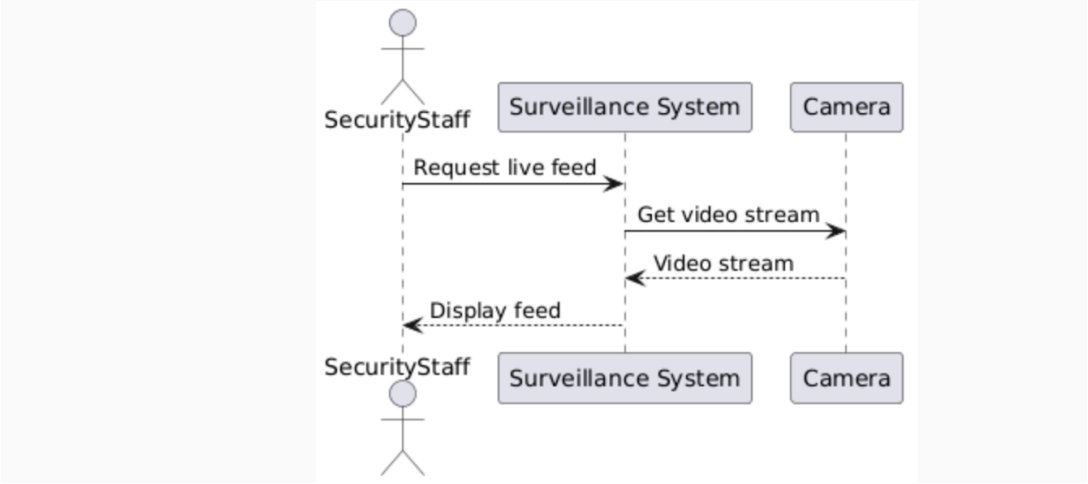

UC-06: Manage Flight Schedule - Sequence Diagram (Refined)
Operations Staff
System UI
Scheduling Controller
Schedule Service
Database
Log in
authenticate()
verifyCredentials()
valid
login success
Open scheduling module
requestModule()
initModule()
module ready
alt
[Create Schedule]
Enter new flight details
submitDetails(data)
createSchedule(data)
[Update Schedule]
Modify existing details
submitDetails(data)
updateSchedule(data)
validateSchedule()
alt
[Valid schedule]
saveSchedule()
success
success
show confirmation
Flight schedule updated
[Invalid schedule]
validation error
show error
Reject schedule
Operations Staff
System UI
Scheduling Controller
Schedule Service
Database


UC-08: Monitor Flight Status - Sequence Diagram
Operations Staff
System UI
Monitoring Controller
Flight Service
Database
Open monitoring dashboard
requestDashboard()
loadFlights()
fetchFlights()
flight list
flights loaded
display dashboard
View flight status
requestStatus(flightId)
getFlightStatus(flightId)
fetchStatus(flightId)
status data
alt
[Data available]
status data
display status
Show updates
loop
[Track updates]
refresh request
getFlightStatus(flightId)
fetchStatus(flightId)
updated data
updated status
refresh display
[Data unavailable]
no data
notify unavailable
Show error message
Operations Staff
System UI
Monitoring Controller
Flight Service
Database

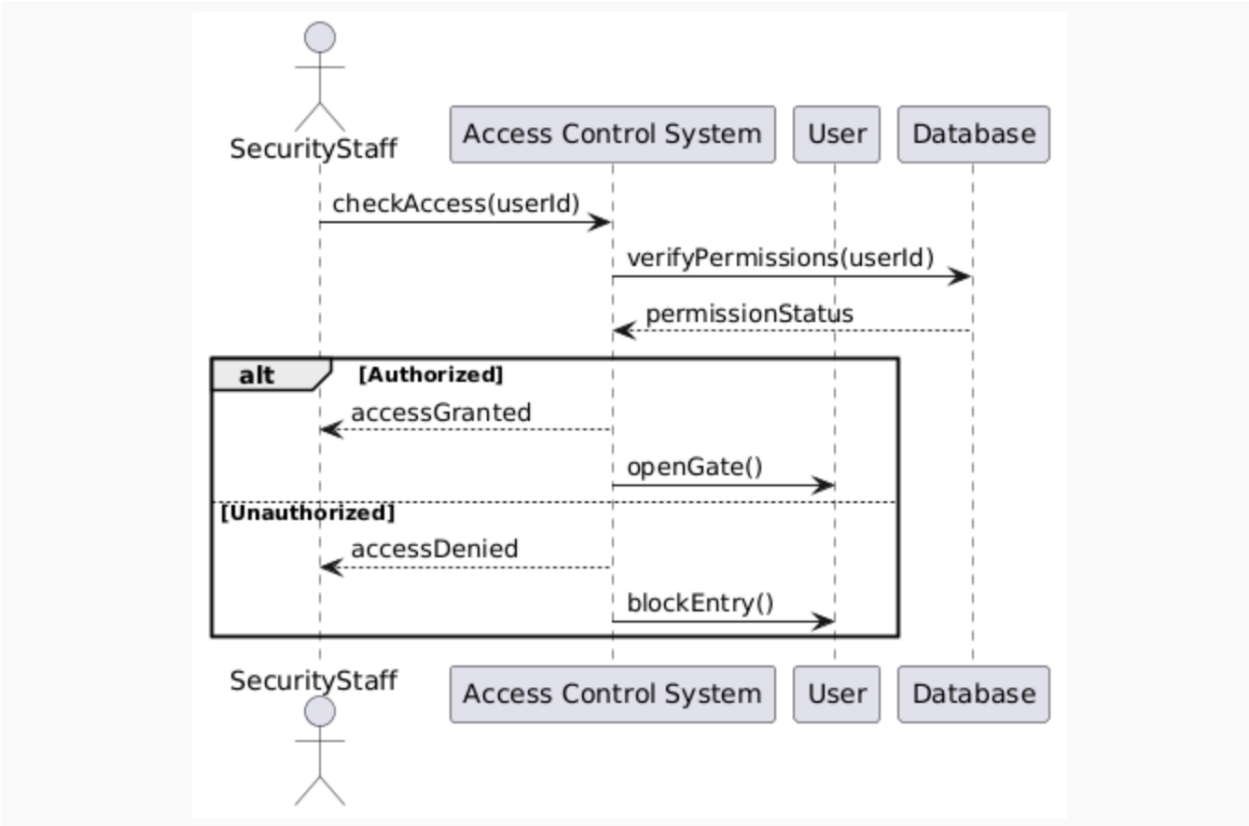

UC-09: Update Operational Records - Sequence Diagram
Operations Staff
System UI
Operations Controller
Record Service
Database
Audit Log
Select record
requestRecord(recordId)
getRecord(recordId)
fetchRecord(recordId)
record data
record data
display record
Modify details
submitUpdates(data)
updateRecord(data)
validateInput()
alt
[Valid input]
saveUpdates(data)
success
writeLog(update action)
update success
show confirmation
Records updated
[Invalid input]
validation error
show error
Error message
Operations Staff
System UI
Operations Controller
Record Service
Database
Audit Log


UC-10: Detect Scheduling Conflicts - Sequence Diagram
Operations Staff
System UI
Scheduling Controller
Conflict Detection Service
Database
Run conflict detection
requestConflictCheck()
detectConflicts()
fetchSchedules()
scheduling data
analyzeSchedules()
alt
[Conflicts found]
conflict list
display conflicts
Show conflicts
[No conflicts]
no conflicts
show confirmation
No conflicts detected
Operations Staff
System UI
Scheduling Controller
Conflict Detection Service
Database

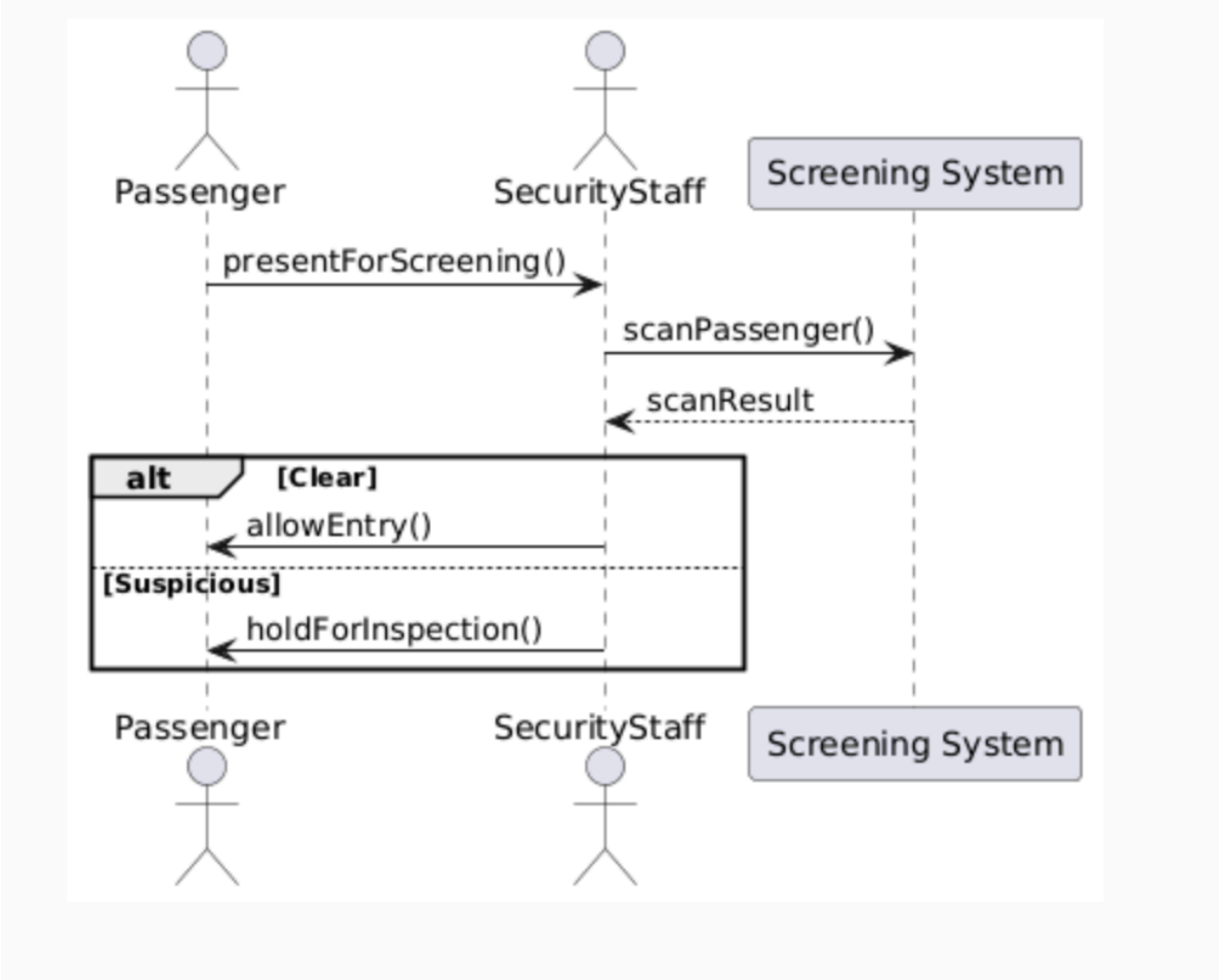
Sequence Diagram UC-11



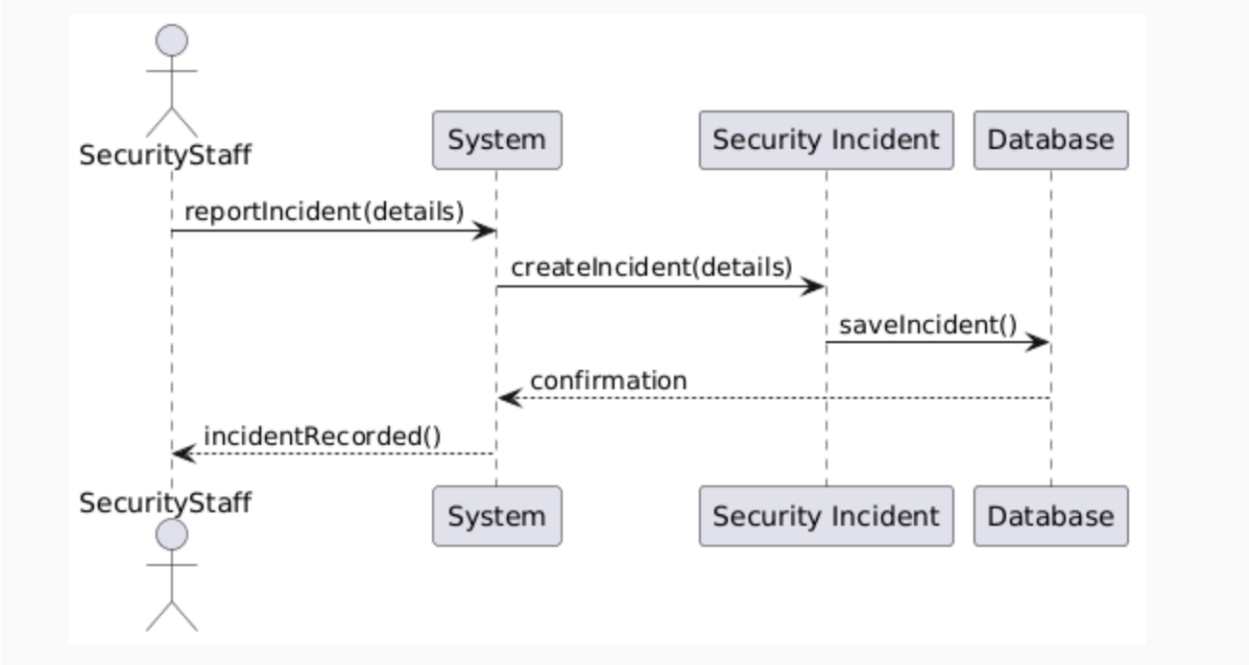
Sequence Diagram UC-12



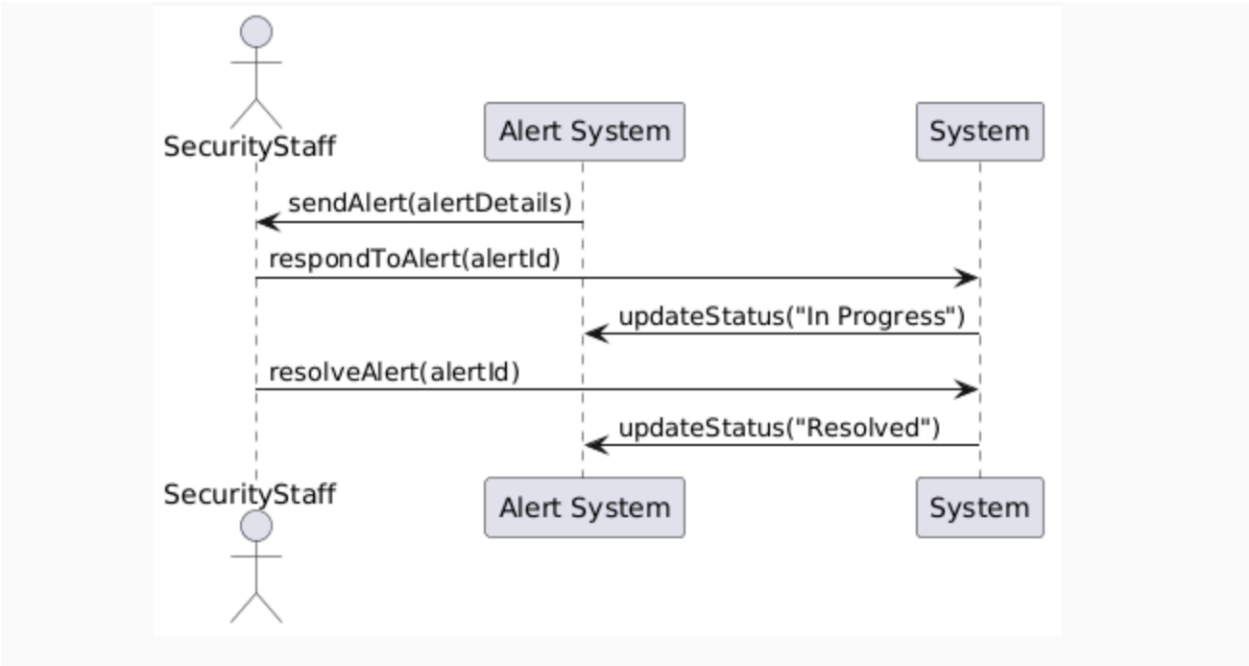
Sequence Diagram UC-13



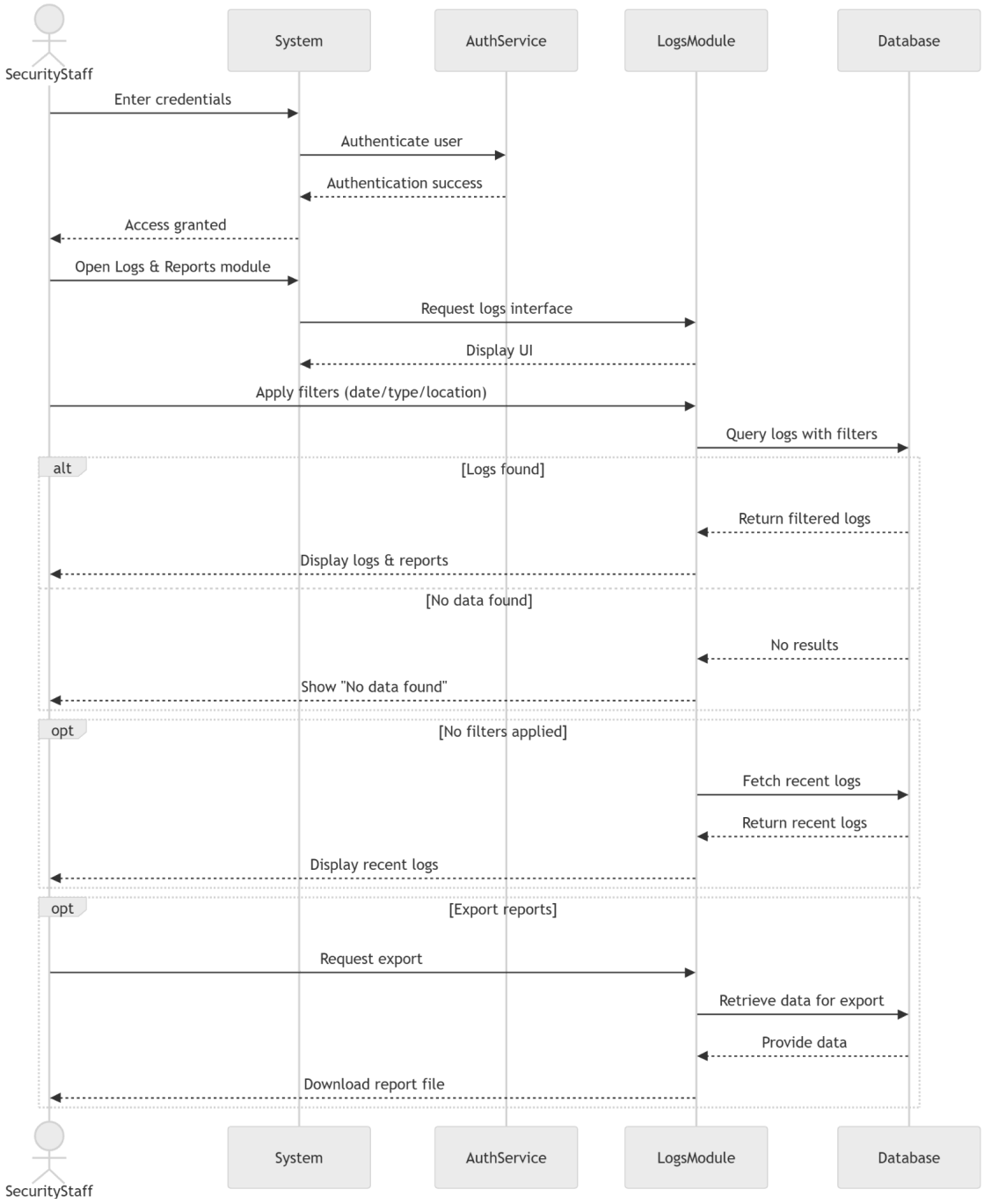
Sequence Diagram UC-14



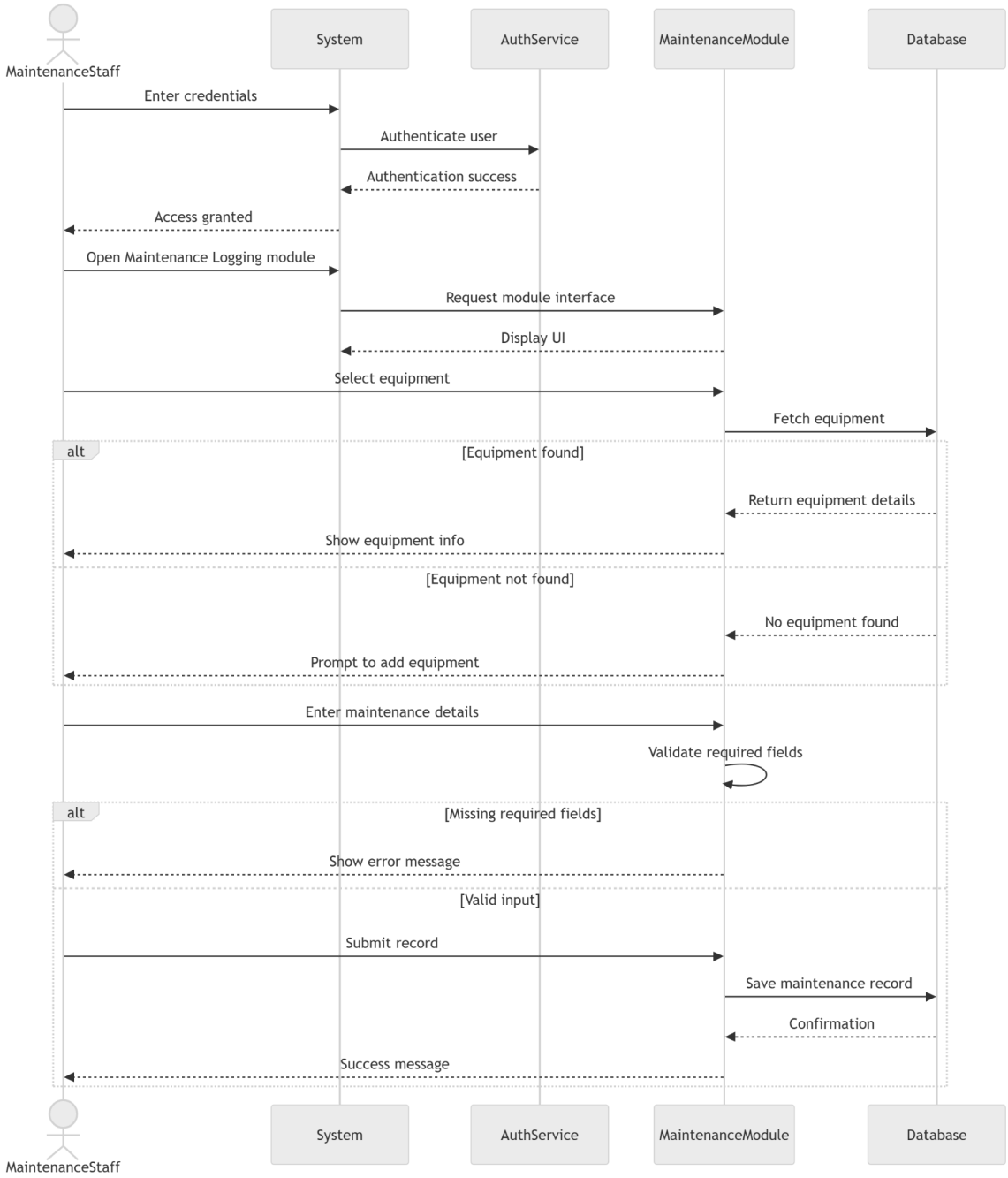
Sequence Diagram UC-15



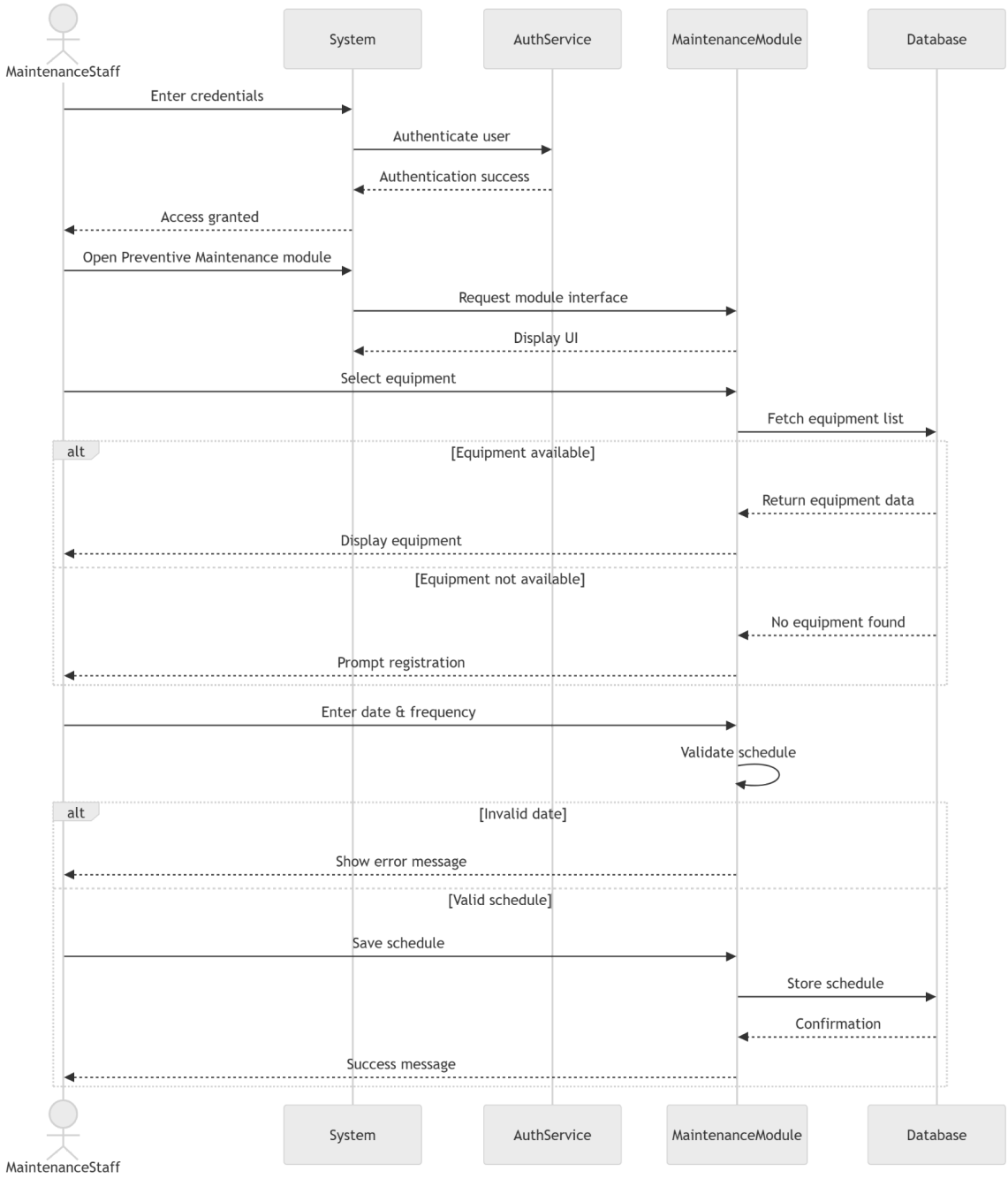
UC-16: Review Security Logs and Reports



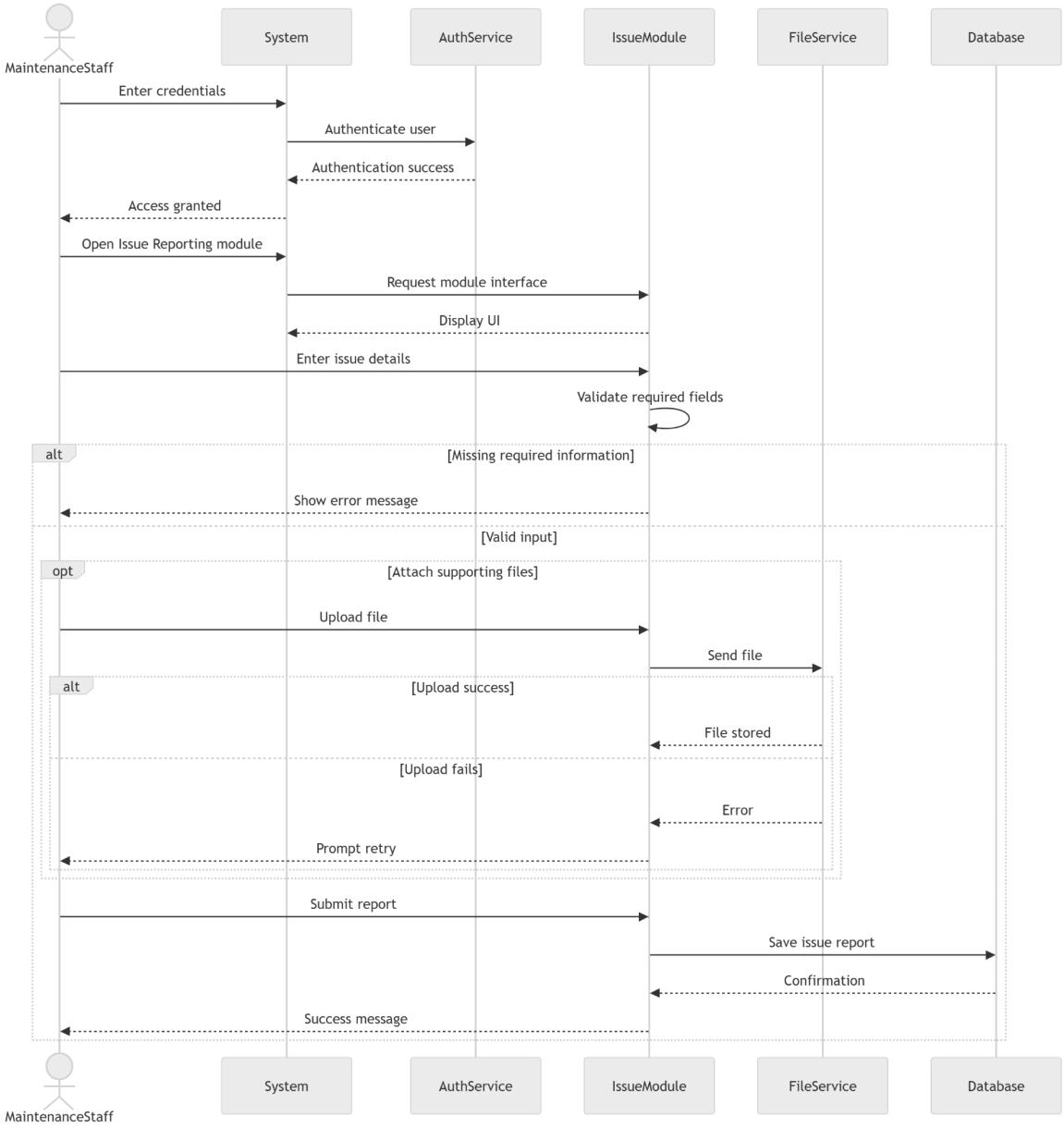
UC-17: Record Maintenance Activities



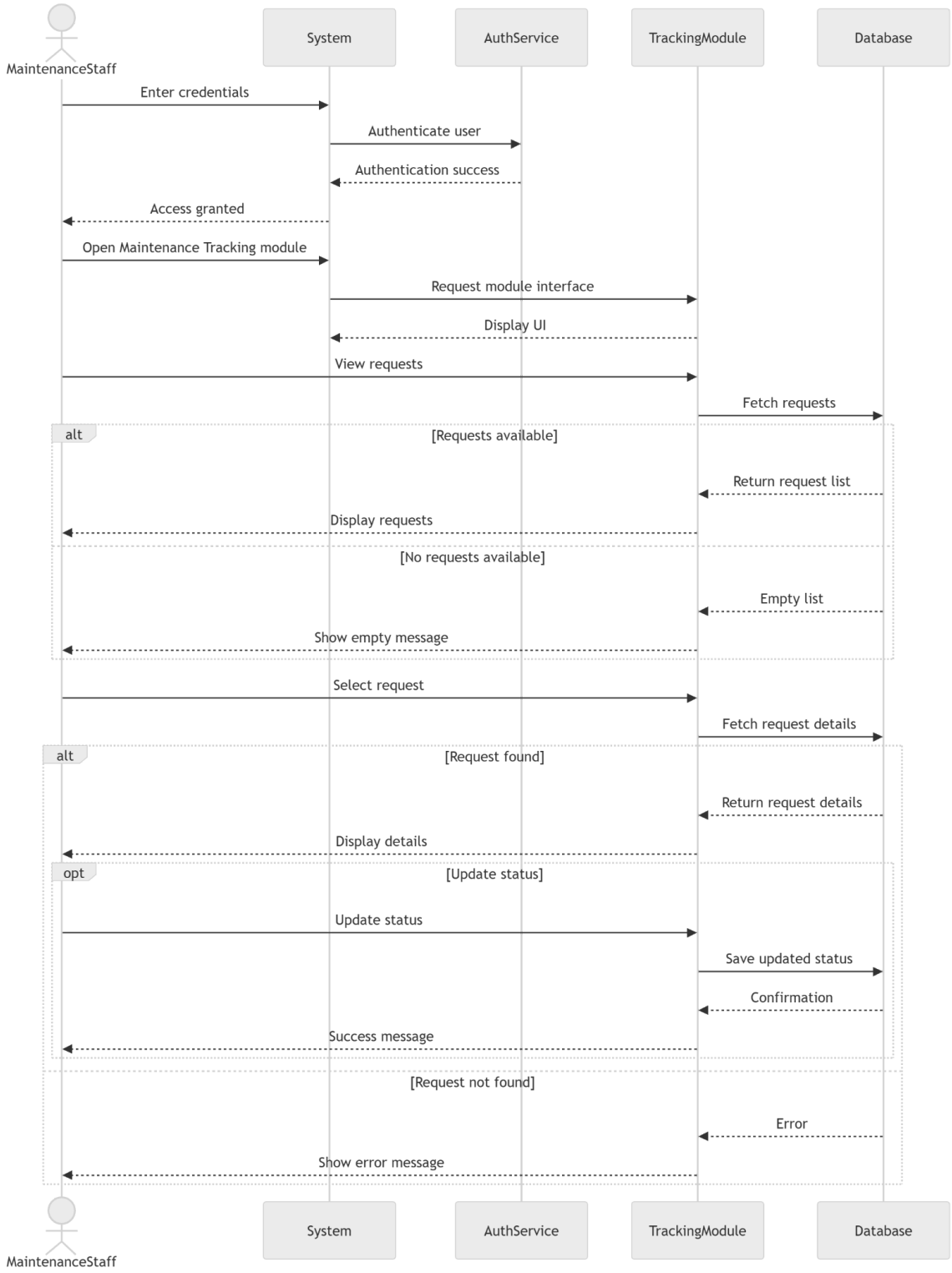
UC-18: Schedule Preventive Maintenance



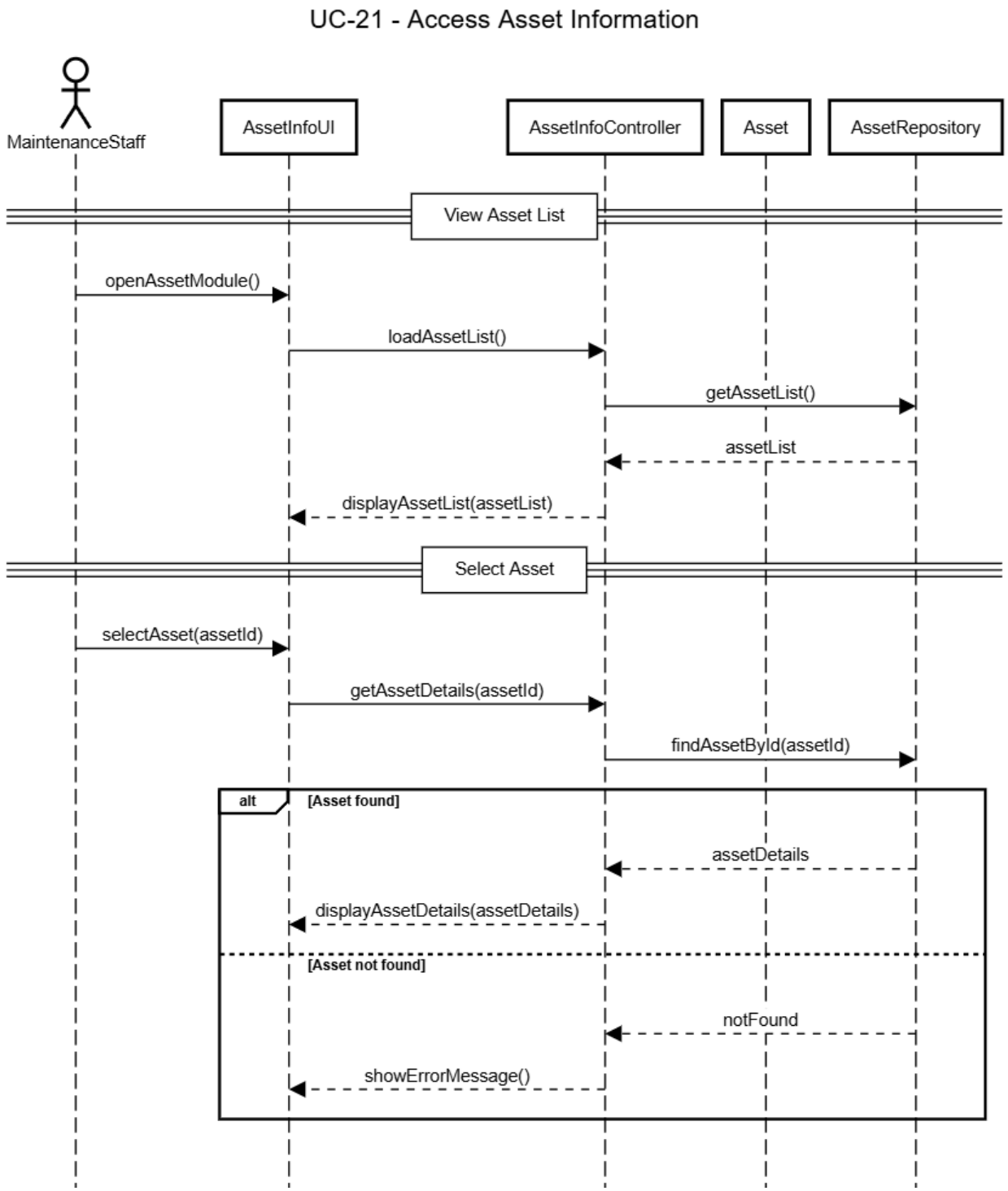
UC-19: Report Infrastructure Issues



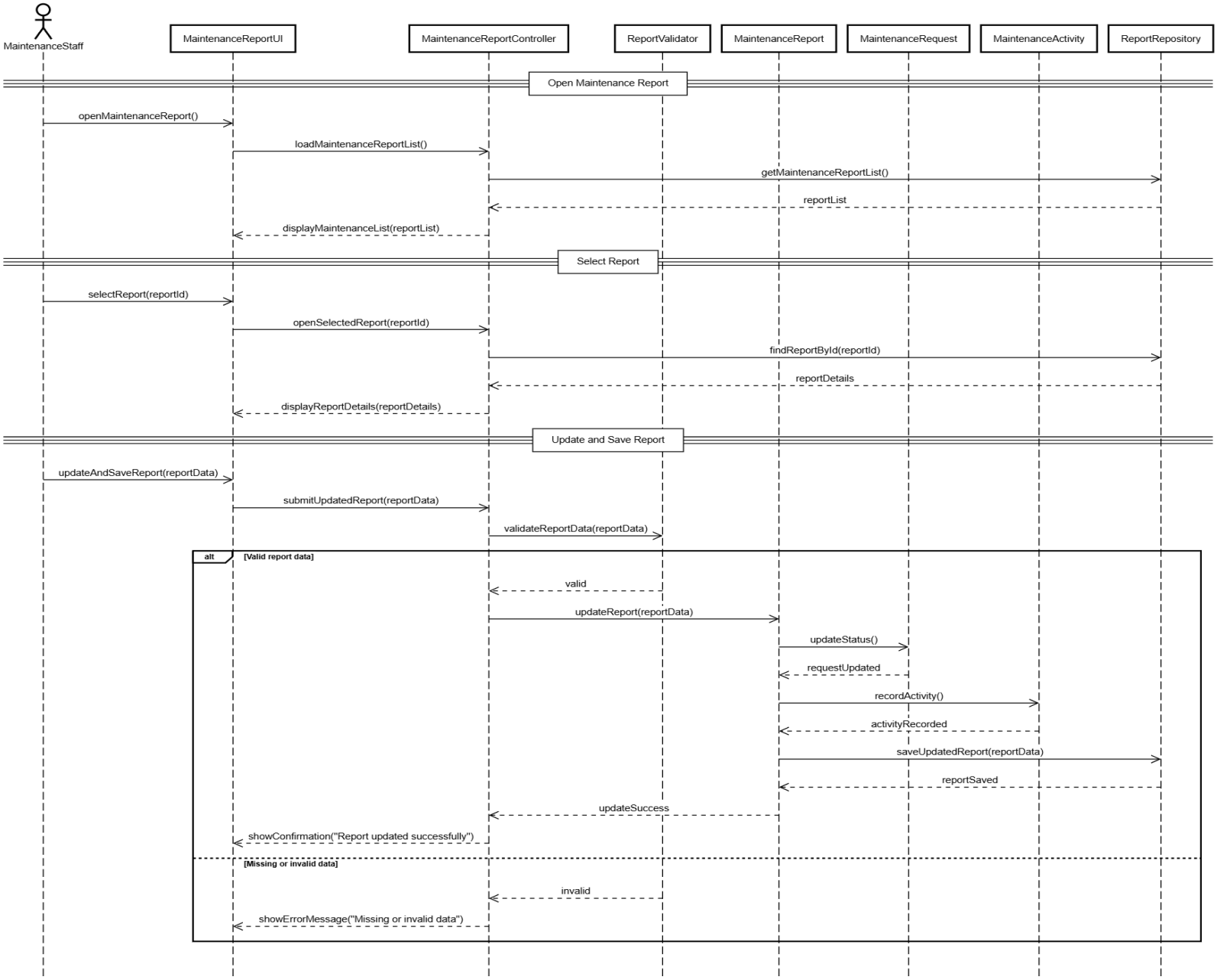
UC-20: Track Maintenance Requests



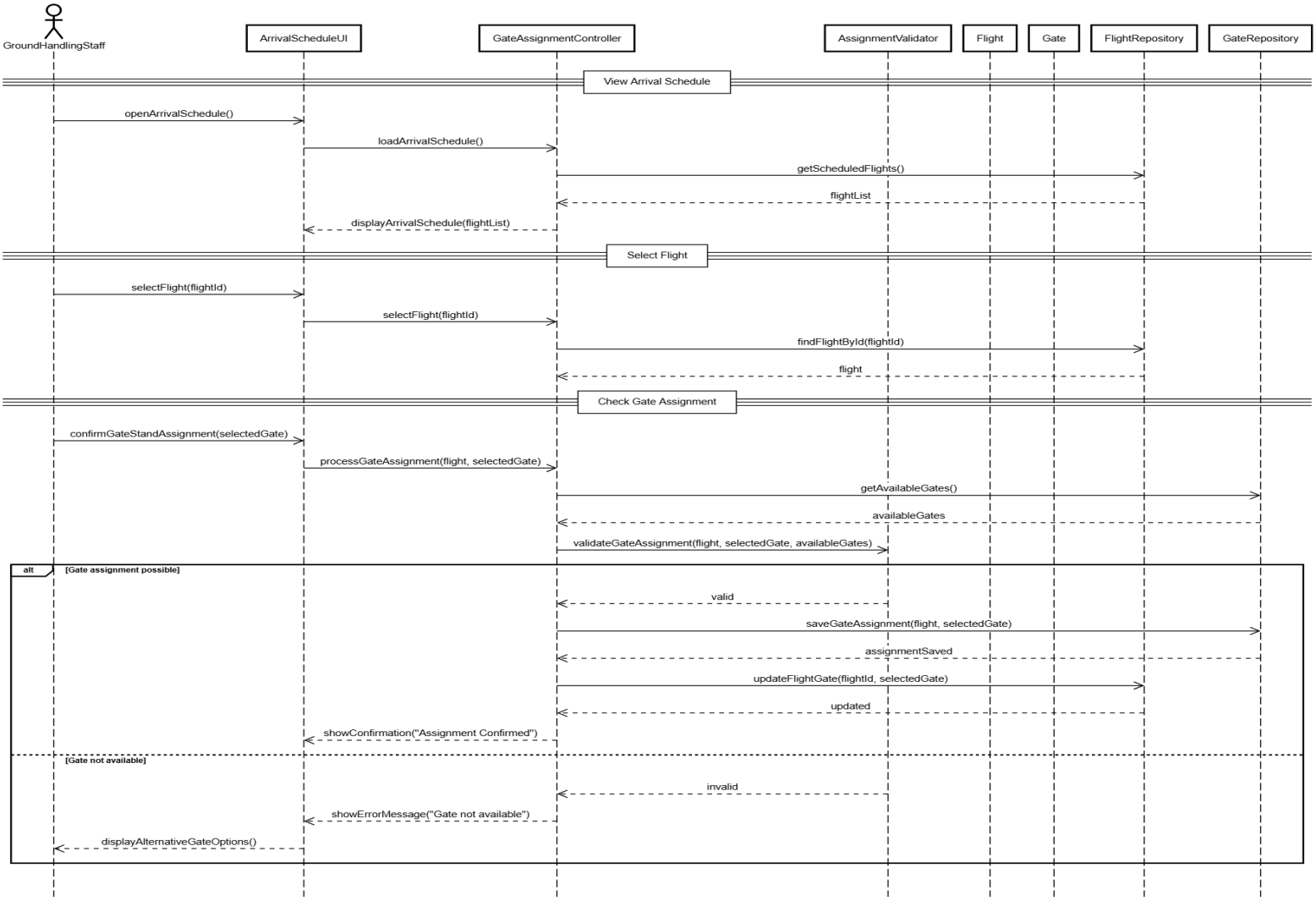
Sequence Diagram



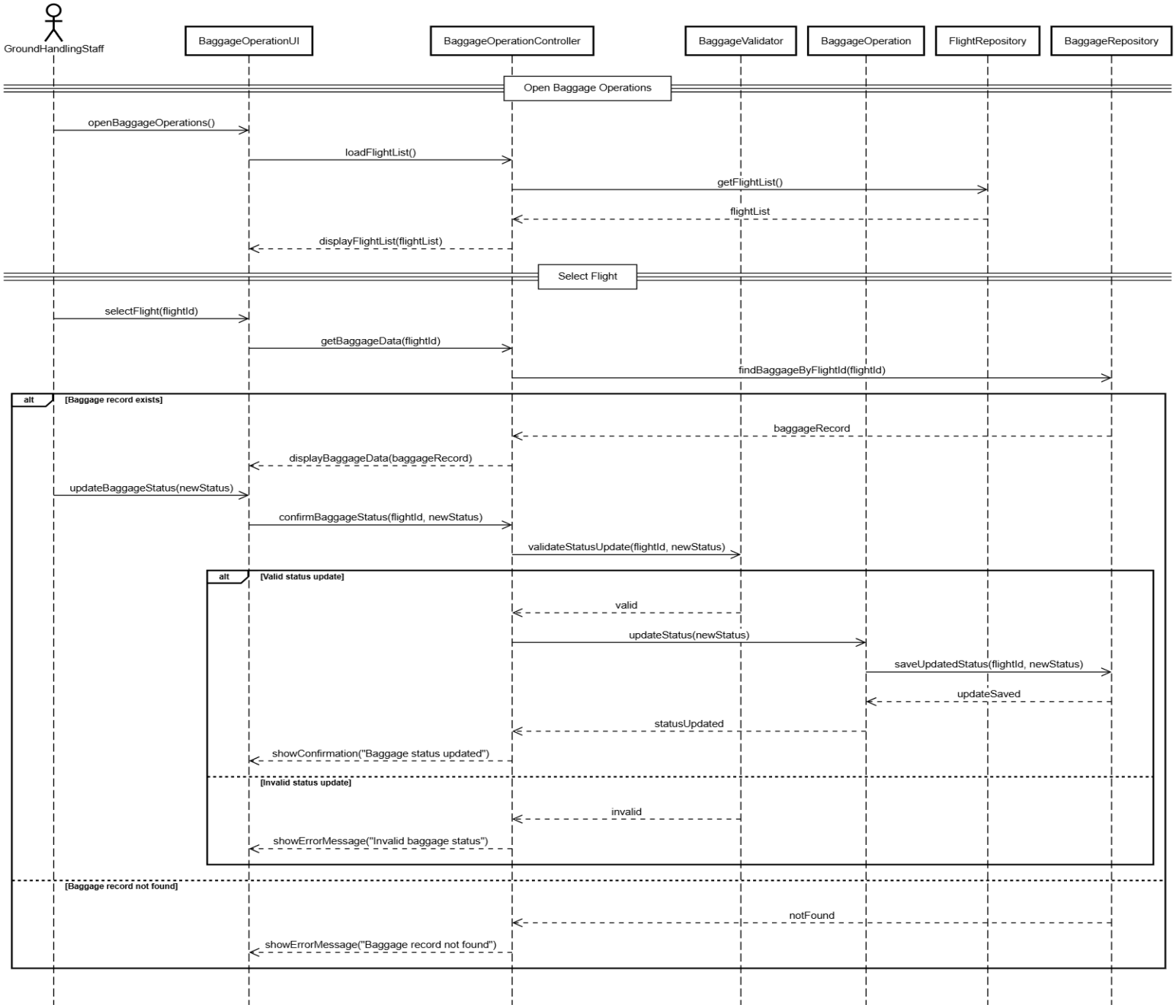
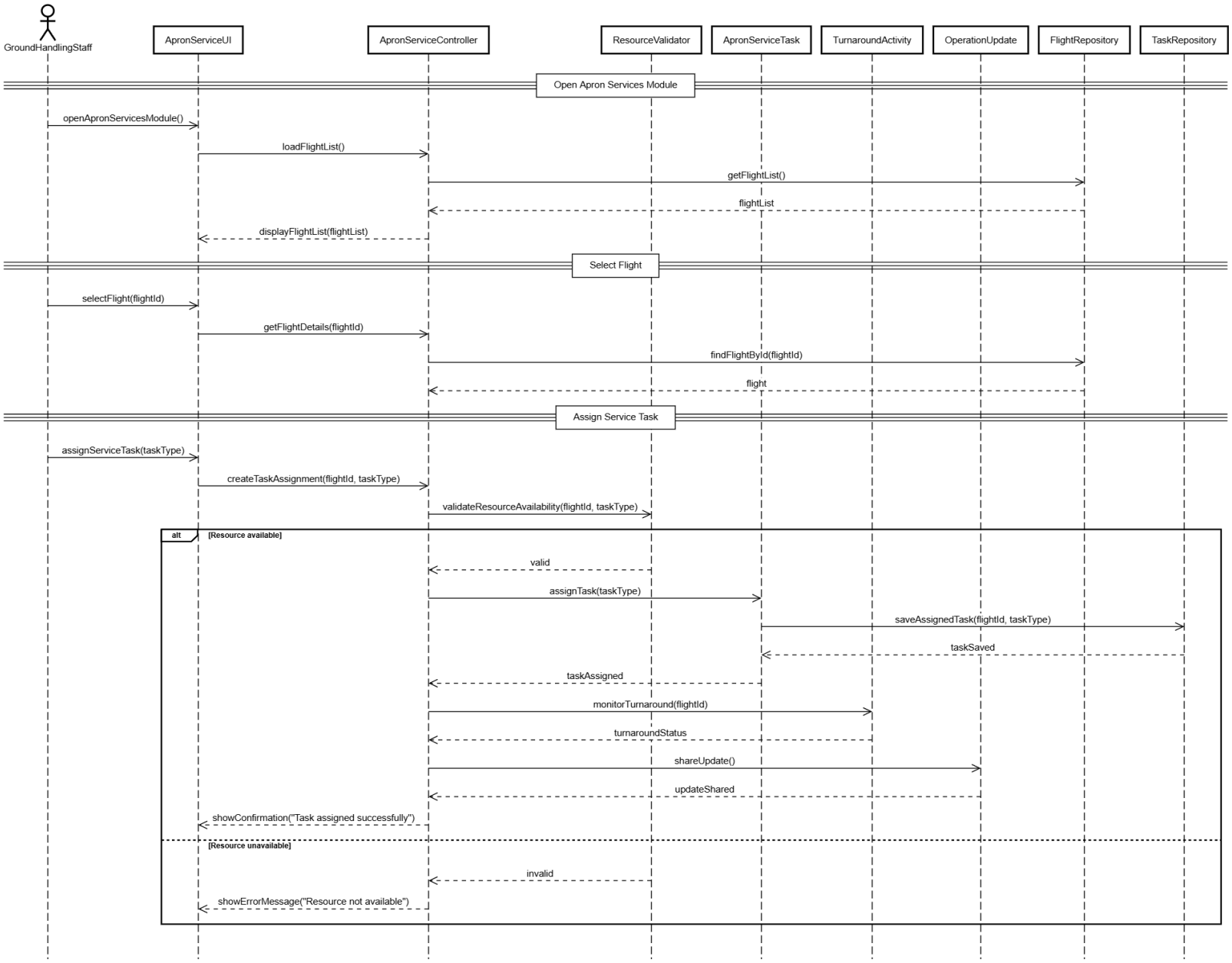
Sequence Diagram UC-22



Sequence Diagram UC-23



Sequence Diagram UC-24



Sequence Diagram UC-25